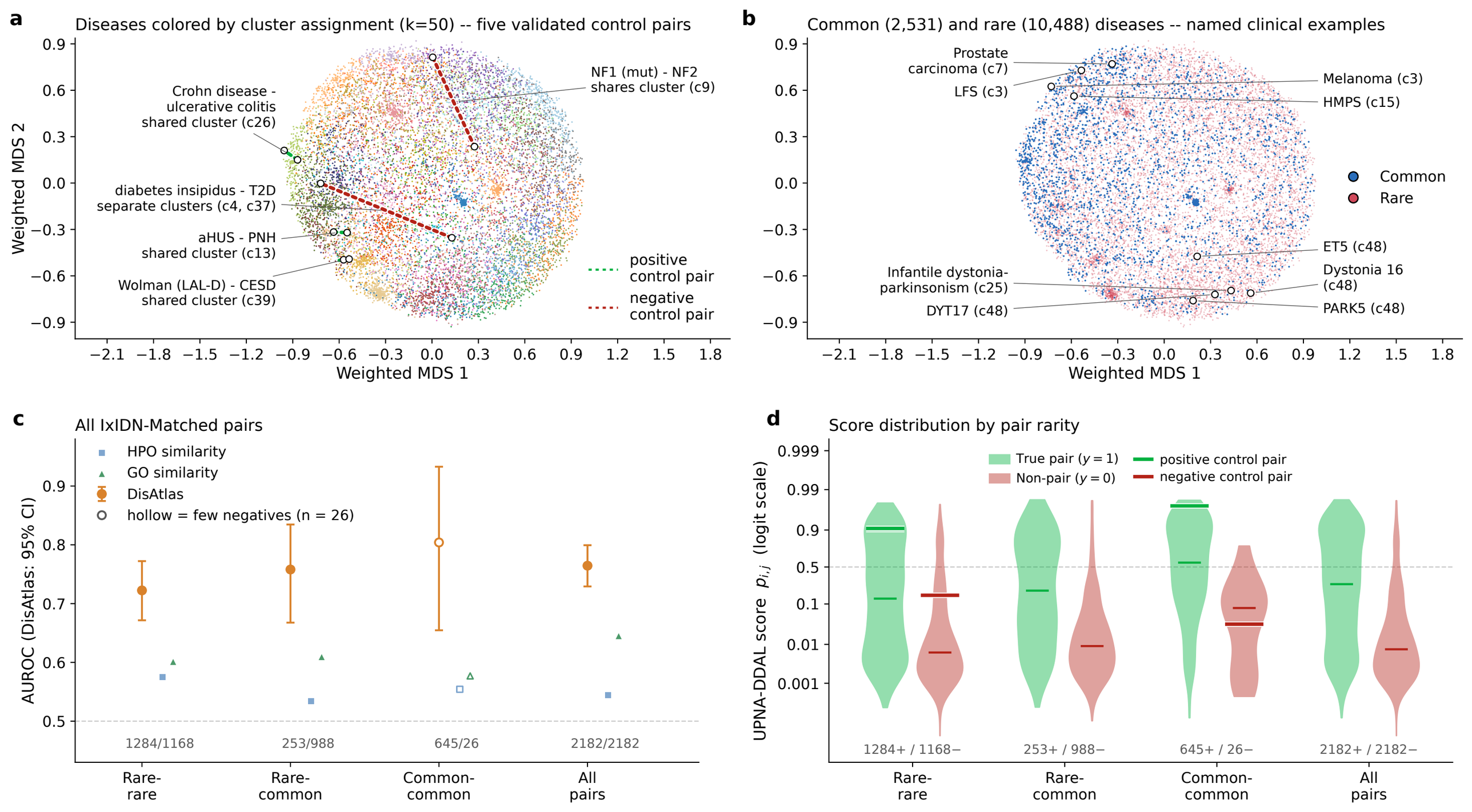

a
Diseases colored by cluster assignment (k=50) -- five validated control pairs
Crohn disease - ulcerative colitis shared cluster (c26)
NF1 (mut) - NF2 shares cluster (c9)
diabetes insipidus - T2D separate clusters (c4, c37)
aHUS - PNH shared cluster (c13)
Wolman (LAL-D) - CESD shared cluster (c39)
positive control pair
negative control pair
Weighted MDS 1
Weighted MDS 2
b
Common (2,531) and rare (10,488) diseases -- named clinical examples
Prostate carcinoma (c7)
LFS (c3)
Melanoma (c3)
HMPS (c15)
Common
Rare
ET5 (c48)
Infantile dystonia-parkinsonism (c25)
Dystonia 16 (c48)
DYT17 (c48)
PARK5 (c48)
Weighted MDS 1
c
All IxIDN-Matched pairs
HPO similarity
GO similarity
DisAtlas
hollow = few negatives (n = 26)
AUROC (DisAtlas: 95% CI)
1284/1168
253/988
645/26
2182/2182
Rare-rare
Rare-common
Common-common
All pairs
d
Score distribution by pair rarity
True pair (y = 1)
Non-pair (y = 0)
positive control pair
negative control pair
UPNA-DDAL score $p_{i,j}$ (logit scale)
1284+ / 1168−
253+ / 988−
645+ / 26−
2182+ / 2182−
Rare-rare
Rare-common
Common-common
All pairs